

MSCI203 Digital Business and Organisational Transformation

Individual assignment

Michaelmas Term 2025/2026

Module leads:

João Baptista (j.baptista@lancaster.ac.uk) / Shiyan Zhang (shiyang.zhang@lancaster.ac.uk)

Individual Assignment – worth 60% of overall module mark

Max word count: 2,500 words. This is a strict limit not a guideline: any piece submitted with more words than the limit will result in the excess not being marked.

AI guidelines: for this assignment you can use AI to refine ideas and research materials but 100% of the text in the submitted assignment must be entirely written and developed by the student. Or in other words, **no text in the assignment can be written by any AI tool.**

In today's workplaces, artificial intelligence (AI) is increasingly introduced to *augment* workers' capabilities. This individual assignment asks you to critically examine the socio-technical dimensions of AI adoption in such contexts and to explore how organizations can manage digital transformation when work is enhanced by AI.

The goal is to assess your ability to engage deeply with course concepts and scholarly literature on digital transformation, AI and socio-technical systems, while structuring a clear and coherent argument.

In your essay, you should address the following questions:

- **AI's Unique Features & Strategic Value:** Describe the unique features of modern AI (e.g. machine learning, generative AI, data-driven decision support) that distinguish it from earlier workplace technologies. Explain how these features enable the augmentation of knowledge work and creative tasks, providing examples of how AI-human collaboration can drive strategic value or innovation in organizations.
- **First-Order and Second-Order Effects:** Examine the immediate impacts (first-order effects) of introducing AI augmentation on workflows, employee roles, and productivity, as well as the longer-term or emergent consequences (second-order effects) on organizational structure and social dynamics. Consider how augmented AI tools might disrupt established formal structures or the informal "social fabric" of the workplace. For example, does AI augmentation change how teams collaborate or make decisions? Does it redistribute expertise and alter power dynamics?
- **Managing the Adoption of AI Augmentation:** Given these socio-technical effects, discuss how organizations should manage the adoption process of AI-augmented work. What strategies can ensure that human-AI collaboration is successful and sustainable? In your discussion, consider the role of leadership and the HR function in change management: for instance, training employees to work effectively with AI, redesigning jobs to balance human and AI strengths, addressing employee concerns (like trust in AI or fear of change), and establishing

policies for ethical and effective use of AI. Your answer should reflect on how to achieve augmentation's benefits while mitigating risks, ensuring that the introduction of AI truly complements human workers rather than creating unintended downsides.

Please structure your essay clearly according to the tasks above and use headings and subheadings to signpost each section of your argument.

Guidance and Expectations:

- Leverage ideas, models and frameworks from the module to frame your analysis. Demonstrate your understanding of these concepts by applying them to the context of AI augmentation in the workplace within organisations.
- Group your argument in literature and show evidence. We expect you to reference **at least 10–15 relevant academic sources**. Prioritize recent research and top-tier literature on AI in organizations, digital workplaces and human-AI collaboration. Higher-quality essays will integrate these references to bolster the analysis (not just describe them) and will also include pertinent **industry examples or case studies** to illustrate key points. Your grade will reflect how well you use these references meaningfully to support the argument and ideas presented.
- Use examples and data from the lecture speakers and group assignment to support your analysis and include relevant outputs (e.g., graphs) to evidence company performance.
- Use your own research and notes to explain and appraise decisions.
- Aim to go beyond a purely descriptive essay. **Critically evaluate** the opportunities and challenges of AI augmentation. For example, if you mention a benefit such as improved decision-making, consider potential counterpoints or limitations.
- Present your ideas in a well-organized manner. Avoid general or broad coverage of ideas. Use introduction and conclusion sections to frame your essay and summarily draw together your argument. Each body section should have a clear focus relating to the questions, and transitions should make the overall logic easy to follow. Writing should be clear and concise – avoid jargon or define it when first used.
- Please be aware that all essays will be checked against plagiarism so be very careful to cite your sources properly. We recommend the use of referencing software such as Endnote or Mendeley to manage in-text citations. You are welcome to include diagrams and figures, screenshots, tables, or any other visual aids to support your argument, but please check guidelines for adding visuals and appendixes to support your argument.

Suggested Academic Readings

To help you get started, below are some recent top-tier publications related to AI in digital workplaces, human-AI collaboration, and socio-technical aspects of augmentation. These are not an exhaustive list, but they offer useful insights and frameworks that you may find relevant:

- **Fuegener et al. (2021, MIS Quarterly):** *“Will Humans-in-the-Loop Become Borgs? Merits and Pitfalls of Working with AI.”* – A multi-method study on human–AI decision-making, finding that while AI advice can improve individual task accuracy, it may also cause workers’ decisions to converge and **reduce unique human knowledge**
- **Abdel-Karim et al. (2022, MIS Quarterly):** *“How AI-Based Systems Can Induce Reflections: The Case of AI-Augmented Diagnostic Work.”* – An empirical study of doctors using an AI decision support tool. It shows that AI systems can prompt deeper **reflective practice** in professionals (a positive augmentation effect), contrasting with purely efficiency-focused views of AI tools.

- **Anthony et al. (2023, Organization Science):** *“‘Collaborating’ with AI: Taking a System View to Explore the Future of Work.”* – A conceptual article that applies socio-technical and organizational theory to AI. It argues that we should view AI not just as a tool, but as a **counterpart in a work system**, and calls for studying how AI’s design, implementation and use are intertwined with work practices.
- **Raisch & Krakowski (2021, Academy of Management Review):** *“Artificial Intelligence and Management: The Automation–Augmentation Paradox.”* – A theoretical piece discussing the intertwined nature of AI automation and augmentation. It posits that augmentation (AI complementing humans) cannot be **neatly separated** from automation; over-emphasizing one side can create paradoxical tensions and unintended consequences.
- **Jia et al. (2024, Academy of Management Journal):** *“When and How Artificial Intelligence Augments Employee Creativity.”* – A mixed-method field study (including an experiment in a company) examining AI assistance in a sales task. It found that AI can boost employees’ creative performance (by taking over routine parts of work so humans focus on complex tasks), **but** the benefits were **“skill-biased,” favoring experts** over less-skilled workers.
- **Shao et al. (2024, Journal of Management):** *“Using Augmentation-Based AI Tool at Work: A Daily Investigation of Learning-Based Benefit and Challenge.”* – A diary study of employees’ daily interactions with an AI tool. It highlights a dual effect: frequent use of an AI assistant led to **knowledge gains and better performance**, yet also caused **information overload** that impaired recovery and performance by day’s end.

Use the references covered in the module and the ones above as a starting point for your own research into different aspects of AI in the workplace. Be sure to integrate insights from such literature to support your arguments on AI augmentation and socio-technical transformation in organizations.

Reminder that the aim of this assignment is to show understanding of key concepts and ability to apply them in developing your own argument. There is no single way to approach the question, we are looking for knowledge and understanding of the concepts. The process of writing a strong essay typically requires several drafts and continuous improvement of your manuscript. Please bear this in mind and allow appropriate time to review and improve the structure and presentation of the materials and ideas. A good way to improve the quality of your essay is to ask colleagues for feedback.

Please adhere to the following formatting guidelines:

- Font: Calibri (body), size 12. With justified paragraphs.
- Line spacing: please use 1.5 line spacing.
- Use of clear headings to separate the discussion of the different functions, and use clear sub-headings to organise different ideas and organise them in a logical way. This helps signpost to the reader the location of the various elements of your work and help navigating the essay.
- Clearly label figures with informative titles, large enough to clearly read, figures should be clearly cited in the text. You should include a selection of important figures within the main body of the report to clearly evidence your discussion. Additional relevant but supporting diagrams and data to be included in an appendix but please do not overdo this.

Harvard referencing must be used throughout.

Good luck!

SUBMISSION DEADLINE: 20:00 (UK time) Monday, 26th January 2026 (TBC)

IMPORTANT NOTES FOR SUBMISSION:

- The deadline is precise, please plan in advance and anticipate the submission
- Please upload a PDF version of the file, please keep it anonymous, do not use your name.
- **All submissions should include a statement that confirms that the content is original and that you have written all the materials yourself and no parts of the assignment were AI generated, and that all sources are correctly and properly acknowledged.**